

HARANADH BABJI PUTTI

Location: Los Angeles , CA | Email: haranadhbabji@gmail.com | Phone: (747) 206-4067 | LinkedIn

SUMMARY

- GIS Analyst with 4+ years experienced in conducting spatial data analysis, geospatial modelling, and utilizing remote sensing technologies.
- Proficient in GIS Tools: ArcGIS (ArcMap, ArcPro), QGIS, Google Earth Engine, PostGIS, FME.
- Strong programming and scripting skills with Python, SQL (PostGIS/MySQL) to automate workflows and perform spatial queries.
- Experienced in Data Visualization using Leaflet, Map box, Power BI, and Tableau to create dynamic, real-time maps and dashboards.
- Adept at managing large-scale geospatial databases using AWS, Google Cloud, PostGIS, and Oracle Spatial for efficient data storage and retrieval.
- Expertise in Remote Sensing technologies, including Lidar data processing, satellite image processing, and NDVI analysis.
- Skilled in Agile/Scrum methodologies for efficient project management and cross-functional team collaboration.
- Strong communication skills with the ability to provide actionable insights and real-time visualizations to stakeholders and decision-makers.

SKILLS

- **Geospatial & GIS Tools:** ArcGIS (ArcMap, ArcPro), ARCGIS Online, QGIS, Google Earth Engine, PostGIS, FME
- **Programming & Scripting:** Python, SQL (PostgreSQL/PostGIS, MySQL), JavaScript (Leaflet)
- **Data Analysis & Spatial Analysis:** Spatial Data Analysis, Geospatial Modelling, Raster and Vector Data Analysis
- **Data Visualization & Mapping:** Cartography, Thematic Mapping, Power BI, Tableau
- **Database Management:** PostGIS, Oracle Spatial, AWS, Google Cloud
- **Remote Sensing:** Satellite Image Processing, Lidar Data Processing, NDVI Analysis
- **GPS Technologies:** GPS Data Collection & Integration, RTK GPS, UAV GIS Integration
- **Project Management:** Agile/Scrum, Jira, Stakeholder Reporting
- **Soft Skills:** Problem-Solving, Collaboration, Time Management, Communication

WORK EXPERIENCE

DXC Technologies

January 2024 – Present

GIS Analyst

- Spearheaded GIS initiatives with ArcGIS Pro and PostGIS to analyse spatial data and optimize traffic flow, reducing congestion in urban areas.
- Carried out geospatial modelling and traffic flow analysis by incorporating satellite and Lidar data to evaluate and optimize road networks.
- Designed dynamic traffic flow maps using Leaflet and Map box, enabling real-time visualizations for stakeholders and city planning teams.
- Employed Python and SQL (PostGIS) for automating traffic data processing workflows and performing advanced spatial queries to analyse traffic patterns.
- Increased data processing efficiency by 25% by automating traffic data workflows and implementing advanced spatial analysis techniques.
- Implemented remote sensing technologies, including Lidar and satellite image processing, to collect high-resolution data for traffic and infrastructure analysis.
- Collaborated with cross-functional teams using Agile methodologies and Jira, tracking progress and ensuring efficient communication and collaboration on project milestones.
- Provided real-time traffic flow insights through interactive dashboards, improving decision-making for urban traffic management and infrastructure development.

Los Angeles Unified School District

February 2024 - December 2024

GIS specialist- Intern

- Developed new map products and modified existing ones using ESRI's ArcMap and ArcGIS Pro to enhance accuracy and ensure compliance with project specifications.
- Performed advanced techniques for labelling, geoprocessing, and creating feature link annotation layers.
- Generated education level maps (elementary, middle, and high school maps) for the district.
- Contributed 55% to the migration of the entire databases and maps from ArcMap to ArcGIS Pro, facilitating a smooth transition.
- Applied advanced mapping concepts and technology trends in GIS cartographic mapping.
- Conducted quality assurance (QA) and quality control (QC) checks on spatial data and map products, ensuring data accuracy, consistency, and adherence to district standards.
- Created custom maps requested by divisional partners that required developing strategies to best convey the map's content to the requestor.
- Assisted with updating and maintaining LAUSD's spatial data, including geodatabases, feature classes, shapefiles, attribute tables, and feature linked annotation layers.

GIS Intern

- Mined U.S. census tract data and integrated it with local and regional transportation datasets to conduct demographic analysis focused on the social equity of transit networks in the Los Angeles area.
- Developed multilingual maps based on Limited English Proficiency (LEP) populations in census tracts for Culver City, aiding in more inclusive urban planning.
- Recreated missing shapefiles from graphic renderings and delivered them to clients for future use.
- Collaborated with city planners and transportation agencies to provide spatial analysis insights that influenced decision-making for transportation projects.

Infosys

April 2021 - July 2022

GIS Analyst

- Led the GIS implementation for urban flood risk analysis, utilizing QGIS and FME to analyse flood-prone areas and assess the potential impact of flooding in urban regions.
- Executed spatial analysis and geospatial modelling to identify flood risk zones, integrating satellite images with environmental data for more accurate flood predictions.
- Used Python to streamline and automate data processing tasks, improving the performance of spatial data analysis and model output generation.
- Leveraged SQL (MySQL) to handle large-scale geospatial databases, optimizing data storage, retrieval, and analysis for flood risk assessments.
- Integrated remote sensing data for satellite image processing and NDVI analysis to monitor environmental changes and urban development that contribute to flood risks.
- Built interactive web maps using Leaflet and Map box, offering stakeholders real-time visualizations of flood risks and affected areas.
- Improved data processing efficiency by 25% through automation and optimization of spatial data workflows.
- Implemented RTK GPS for precise data collection and integration, ensuring the accuracy of spatial data used for flood mapping and risk analysis.

Trigent Software

January 2020 - March 2021

GIS Analyst

- Managed the GIS integration for urban planning, focusing on infrastructure management with ArcGIS (ArcMap) and PostGIS.
- Conducted spatial data analysis and geospatial modelling with PostGIS, optimizing mapping of urban resources and utilities for city development.
- Utilized Python and R for automating geospatial data analysis tasks and improving the efficiency of data processing workflows.
- Applied SQL (PostgreSQL/PostGIS) for managing and querying large-scale spatial databases, ensuring accurate and timely data retrieval for city planning departments.
- Developed thematic maps and visualized urban growth patterns using ArcGIS, helping municipal authorities make informed decisions about infrastructure and resource allocation.
- Collaborated with cross-functional teams using Agile/Scrum methodologies, improving project timelines and ensuring on-time delivery of project milestones.
- Employed Jira for tracking project progress, managing tasks, and ensuring efficient communication across teams, ensuring alignment with client expectations.
- Increased data processing efficiency by 30% through the automation of spatial data workflows and advanced analysis techniques.

PROJECTS**Smart City Traffic Flow Optimization**

- Led spatial analysis and geospatial modelling with ArcGIS Pro and PostGIS to optimize traffic flow and reduce congestion in urban areas.
- Developed dynamic, interactive traffic maps using Leaflet and Map box, providing real-time traffic visualizations for stakeholders.

Real-Time Environmental Data Dashboard

- Developed real-time environmental data visualizations and dashboards using ArcGIS Online, Tableau, and Power BI.
- Used Python for automating data updates and ensuring timely visualization of environmental parameters.

EDUCATION**Master of Science in Geographic Information Science**

Aug 2022 – May 2025

California State University, Northridge | GPA: 3.8/4.0

Bachelor of Technology in Petroleum Engineering

Jun 2017 – May 2021

Vignan's Foundation for Science, Technology & Research Institute | GPA: 3.25/4.0